PREVENTIVE MAINTENANCE REPORT

Full Technical Assessment & System Health

PREPARED FOR:

Bank of Dream Public Company Limited

PREPARED BY:

[PTS] Professional Team Services

Report Cycle: 2025Q4
Generated Date: 2025-12-19 00:30
Total Hosts Monitored: 3

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DOCUMENT CONTROL & PREFACE

1. Document Information

Report Type:	FULL TECHNICAL REPORT
System Scope:	PM-Core Automation Monitoring
Classification:	Confidential

2. Objective & Scope

This document provides a **comprehensive technical health check analysis** of the computer systems. The objectives are:

• To analyze detailed system resource utilization (CPU, Memory, Disk).
• To verify system availability and identify operational risks.
• To serve as operational evidence for **ISO 20000 / ISO 27001** compliance audits.

3. Contact Information

For any inquiries regarding this report, please contact:
Professional Team Services
Email: contactcenter@g-able.com
Tel: 02-781-9000

Executive Summary

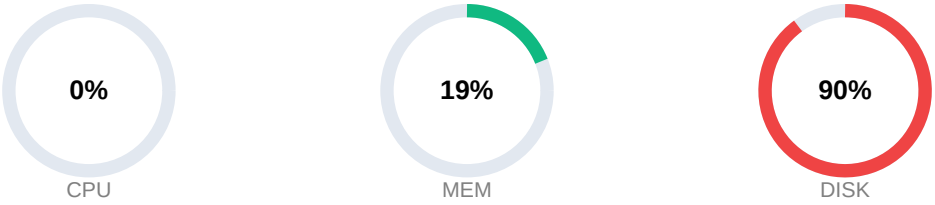
HEALTH SCORE
76%

Healthy Systems	2
Warning Systems	0
Critical Systems	1
Offline / No Data	0

Host Status Overview

Hostname / IP	OS	Last Seen	Status	CPU	MEM	DISK
GSBSERVER 192.168.1.62 192.168.56.103	CentOS Linux 7 (Core)	19 Dec 00:29	Degraded	0%	19%	90%
centserver1.local 192.168.56.101	CentOS Stream 9	19 Dec 00:08	Up	24%	23%	5%
redhatserver1.local 192.168.56.102	Red Hat Enterprise Linux	19 Dec 00:08	Up	53%	31%	10%

Host Detail: GSBSERVER



System Information

Hostname:	GSBSERVER
Data Snapshot:	2025-12-19 00:29:40 (Thai Time)
OS:	CentOS Linux 7 (Core)
Kernel:	3.10.0-1160.el7.x86_64
Uptime:	14 hours, 29 minutes
IP:	192.168.1.62 192.168.56.103
Platform / Virt:	VirtualBox
Host ID:	a8c03e01

Mount Points

Mount	Type	Size	Used	Usage %	Status
/repo/centos79	iso9660	9.5G	9.5G	100.0%	Full/Crit
/	xfs	116G	103G	90.0%	Full/Crit
/boot	xfs	509M	144M	29.0%	OK
/run	tmpfs	1.4G	41M	3.0%	OK
/dev	devtmpfs	1.4G	0	0.0%	OK
/dev/shm	tmpfs	1.4G	0	0.0%	OK
/sys/fs/cgroup	tmpfs	1.4G	0	0.0%	OK
/run/user/1000	tmpfs	285M	0	0.0%	OK

Resource Usage: Top Processes

Top 5 Processes by CPU Usage	%
NetworkManager	3.1%
audispd	1.5%
sshd	1.1%
sshd	1.1%
pm_collect.sh	1.0%

Top 5 Processes by Memory Usage	%
firewalld	1.0%
tuned	0.5%
polkitd	0.5%
NetworkManager	0.3%
lvmetad	0.2%

Repositories

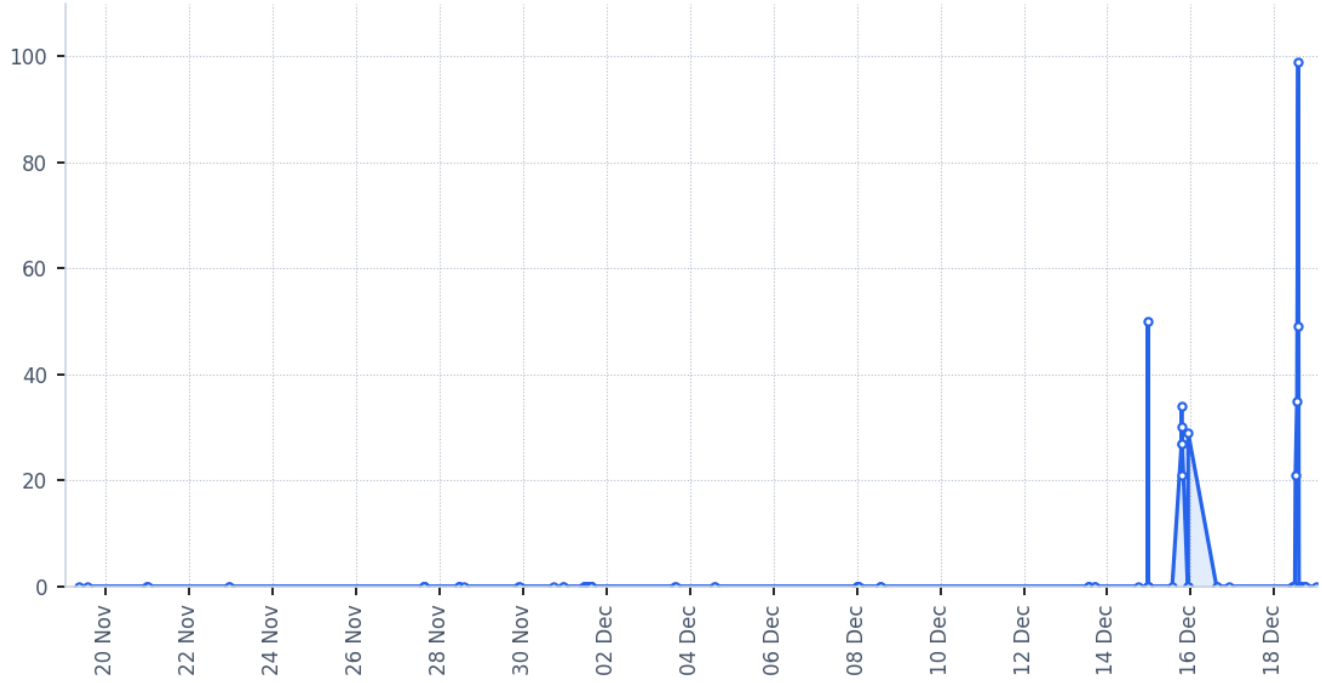
Repo ID	Name	Type
centos79	CentOS 7.9 media 10,072	Local

Analysis & Recommendations

- **Critical** Disk **/repo/centos79** (100.0%). Cleanup required.
- **Critical** Disk **/** (90.0%). Cleanup required.

CPU Performance Analysis: GSBSERVER

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	0.0%	99.0%	3.3%	Transient Peak

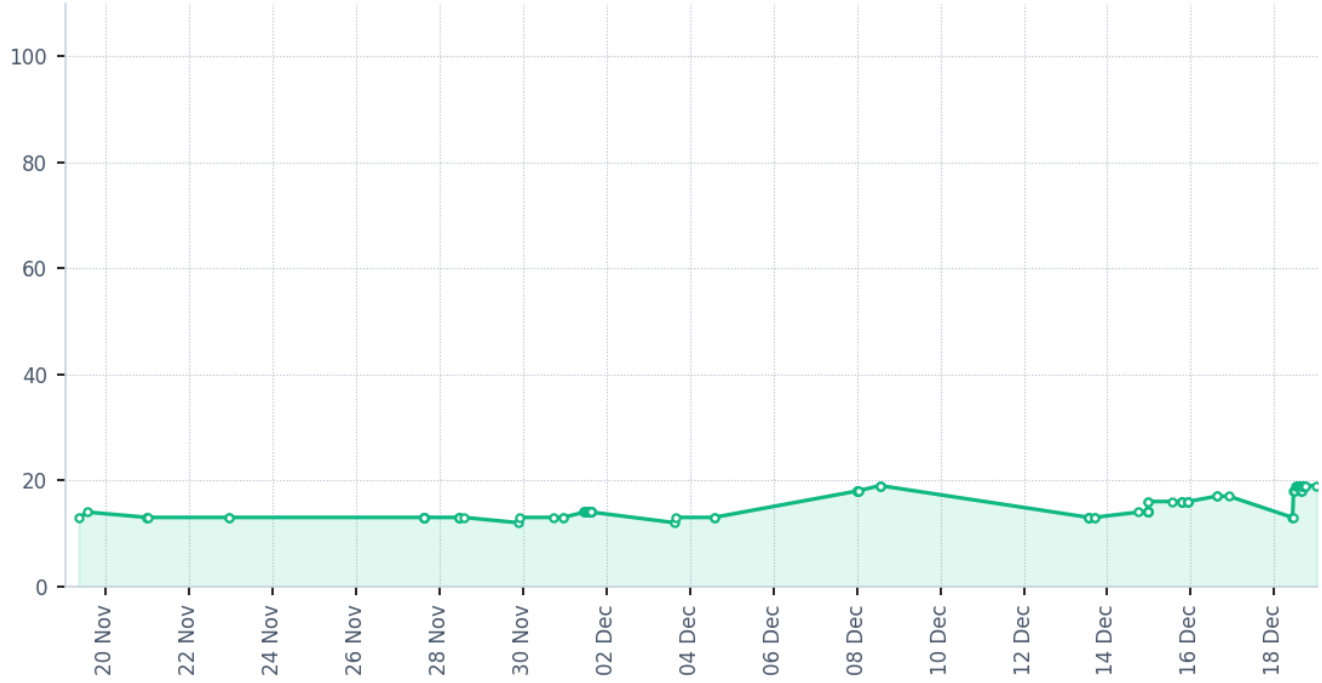
Analysis & Observations

Transient Spikes: While average load is normal (3.3%), critical peaks up to 99.0% were detected. Investigate specific scheduled tasks.

Data consistency check passed. No significant gaps detected in the monitoring period.

Memory Usage Analysis: GSBSERVER

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	19.0%	19.0%	15.3%	Normal
Swap	-	0.0%	-	Normal

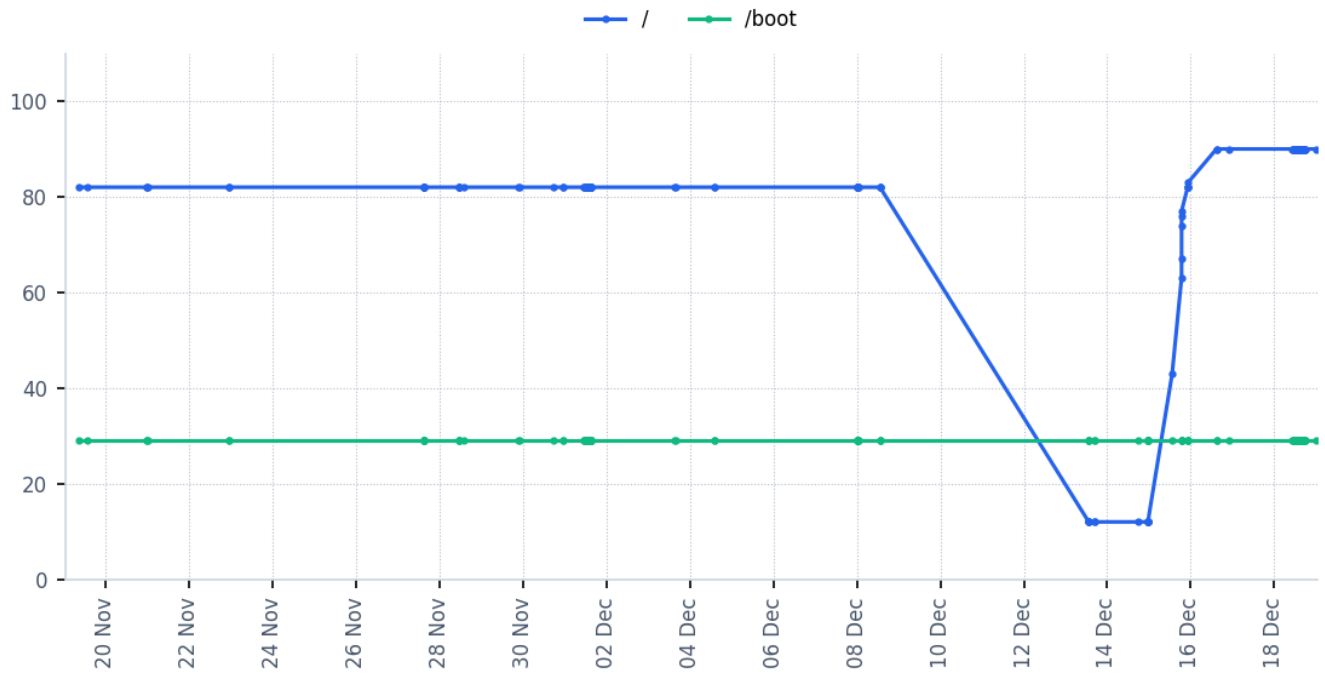
Analysis & Observations

Normal Operation: Resource consumption is within standard baselines (Avg: 15.3%).

Data consistency check passed. No significant gaps detected in the monitoring period.

Disk Usage Analysis (OS Partitions): GSBSERVER

Scope: 19 Nov 2025 - 19 Dec 2025



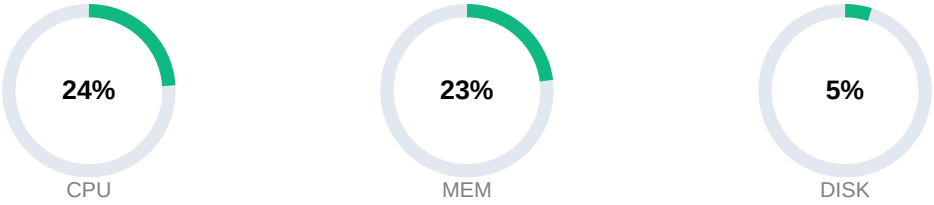
Statistics Summary

Mount Point	Size	Current	Peak (30d)	Assessment
/	116G	90.0%	90.0%	Critical: Space Exhausted
/boot	509M	29.0%	29.0%	Normal Operation

Analysis & Observations

Disk usage analysis is detailed in the table above based on OS partition monitoring.

Host Detail: centserver1.local



System Information

Hostname:	centserver1.local
Data Snapshot:	2025-12-19 00:08:30 (Thai Time)
OS:	CentOS Stream 9
Kernel:	5.14.0-319.el9.x86_64
Uptime:	15 hours, 37 minutes
IP:	192.168.56.101
Platform / Virt:	VirtualBox
Host ID:	a8c06538

Mount Points

Mount	Type	Size	Used	Usage %	Status
/boot	xfs	960M	232M	25.0%	OK
/run	tmpfs	356M	19M	6.0%	OK
/	xfs	27G	1.3G	5.0%	OK
/dev	devtmpfs	4.0M	0	0.0%	OK
/dev/shm	tmpfs	890M	0	0.0%	OK
/run/user/1000	tmpfs	178M	0	0.0%	OK
/run/user/0	tmpfs	178M	0	0.0%	OK

Resource Usage: Top Processes

Top 5 Processes by CPU Usage	%
systemd-logind	7.0%
pm_collect.sh	3.0%
systemd-journal	2.0%
kworker/0:3-eve	1.0%
systemd	0.0%

Top 5 Processes by Memory Usage	%
firewalld	2.3%
NetworkManager	1.1%
systemd	0.8%
systemd	0.7%
systemd	0.7%

Repositories

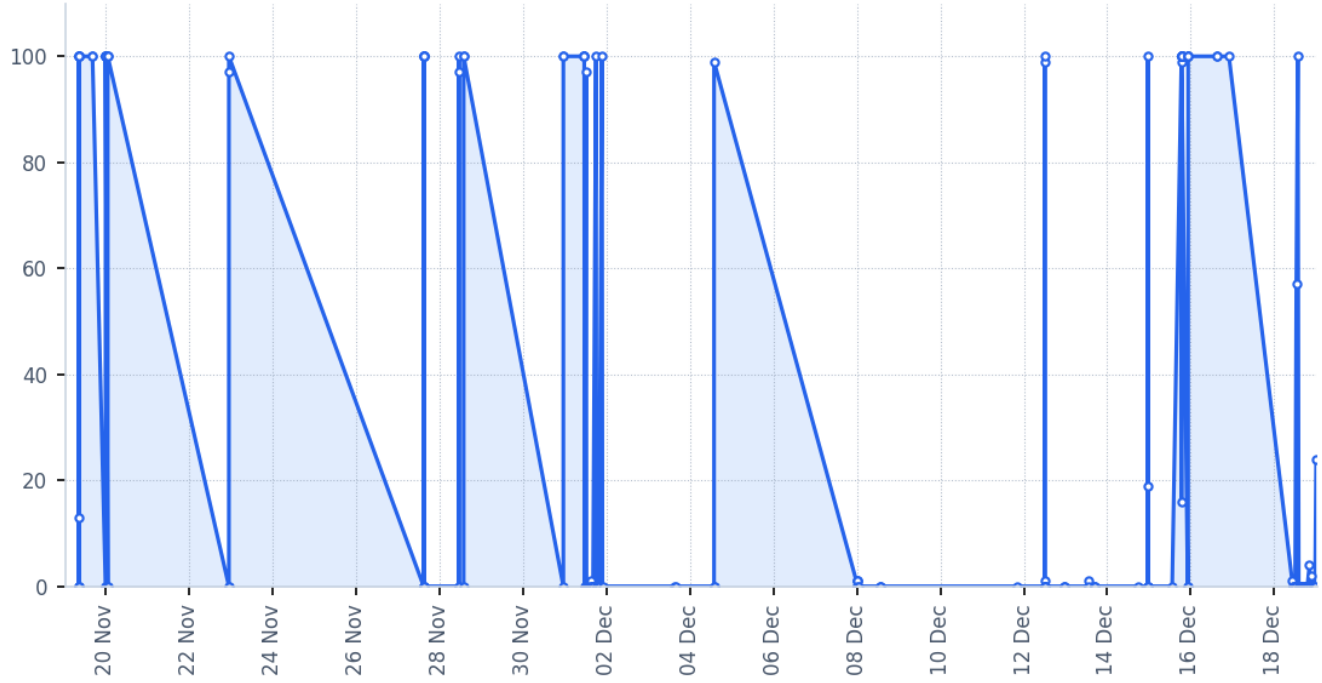
Repo ID	Name	Type
appstream	CentOS Stream 9 - AppStream	Network
baseos	CentOS Stream 9 - BaseOS	Network
extras-common	CentOS Stream 9 - Extras packages	Network

Analysis & Recommendations

- All systems healthy (<80%).

CPU Performance Analysis: centserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	24.0%	100.0%	32.9%	Transient Peak

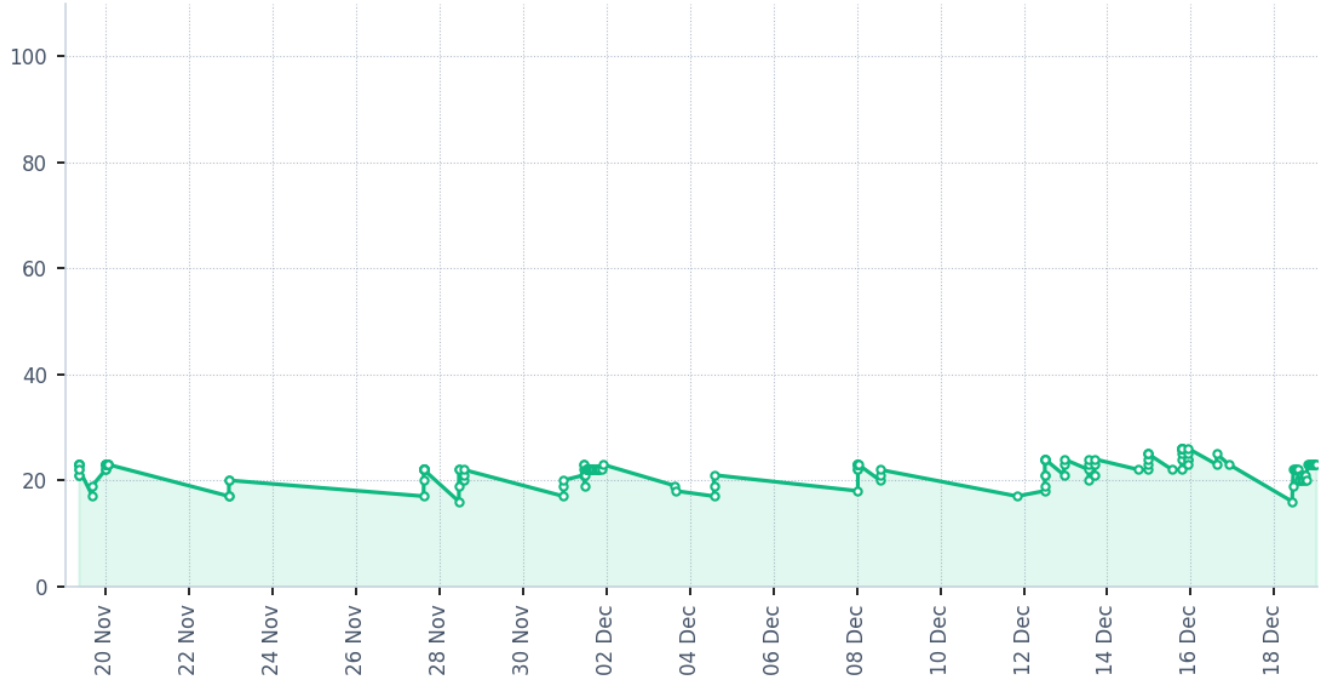
Analysis & Observations

Transient Spikes: While average load is normal (32.9%), critical peaks up to 100.0% were detected. Investigate specific scheduled tasks.

Data consistency check passed. No significant gaps detected in the monitoring period.

Memory Usage Analysis: centserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	23.0%	26.0%	22.0%	Normal
Swap	-	0.0%	-	Normal

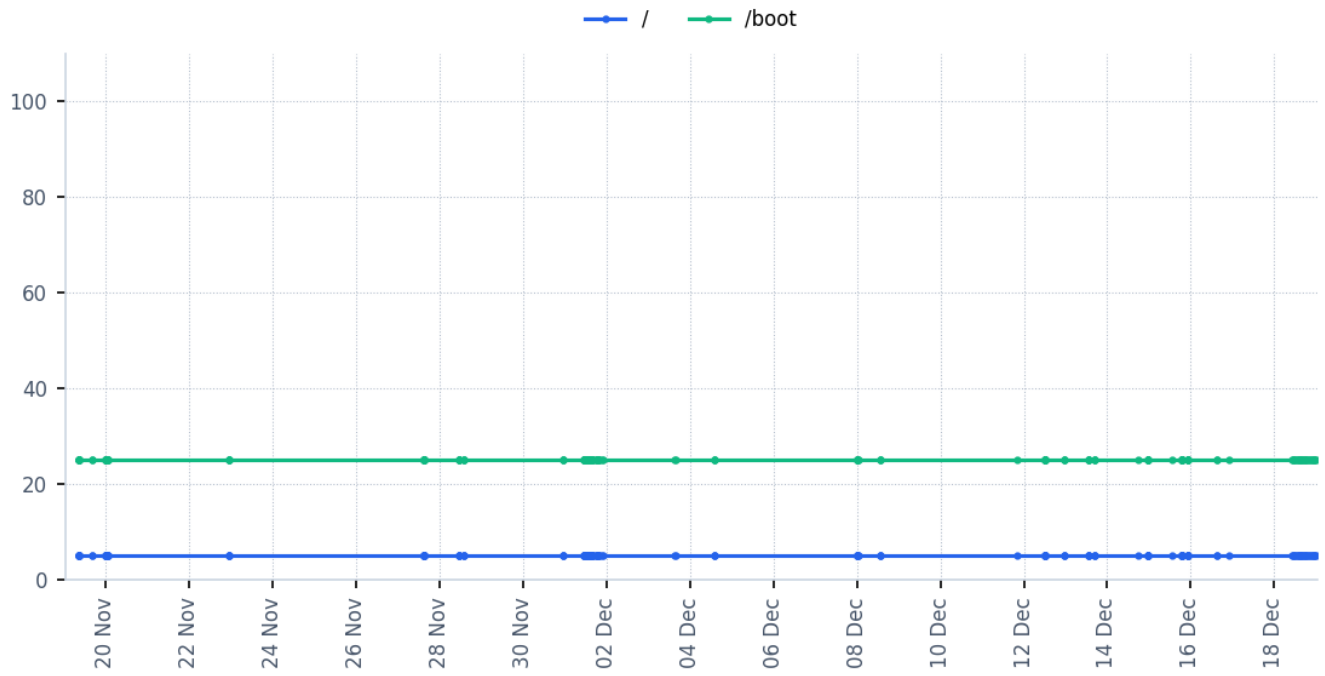
Analysis & Observations

Normal Operation: Resource consumption is within standard baselines (Avg: 22.0%).

Data consistency check passed. No significant gaps detected in the monitoring period.

Disk Usage Analysis (OS Partitions): centserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



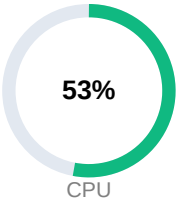
Statistics Summary

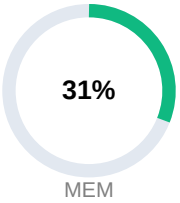
Mount Point	Size	Current	Peak (30d)	Assessment
/	27G	5.0%	5.0%	Normal Operation
/boot	960M	25.0%	25.0%	Normal Operation

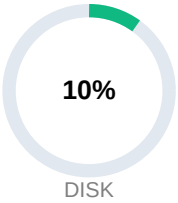
Analysis & Observations

Disk usage analysis is detailed in the table above based on OS partition monitoring.

Host Detail: redhatserver1.local







System Information

Hostname:	redhatserver1.local
Data Snapshot:	2025-12-19 00:08:30 (Thai Time)
OS:	Red Hat Enterprise Linux 9.5 (Plow)
Kernel:	5.14.0-503.11.1.el9_5.x86_64
Uptime:	15 hours, 37 minutes
IP:	192.168.56.102
Platform / Virt:	VirtualBox
Host ID:	a8c06638

Mount Points

Mount	Type	Size	Used	Usage %	Status
/boot	xfs	960M	347M	37.0%	OK
/	xfs	46G	4.6G	10.0%	OK
/run	tmpfs	556M	16M	3.0%	OK
/run/user/42	tmpfs	278M	52K	1.0%	OK
/run/user/1001	tmpfs	278M	36K	1.0%	OK
/run/user/0	tmpfs	278M	36K	1.0%	OK
/dev	devtmpfs	4.0M	0	0.0%	OK
/dev/shm	tmpfs	1.4G	0	0.0%	OK

Resource Usage: Top Processes

Top 5 Processes by CPU Usage	%
systemd-logind	20.0%
systemd-udevd	10.0%
pm_collect.sh	4.0%
systemd	0.0%
kthreadd	0.0%

Top 5 Processes by Memory Usage	%
gnome-shell	8.7%
ibus-x11	1.8%
Xwayland	1.6%
firewalld	1.6%
gsd-color	1.0%

Repositories

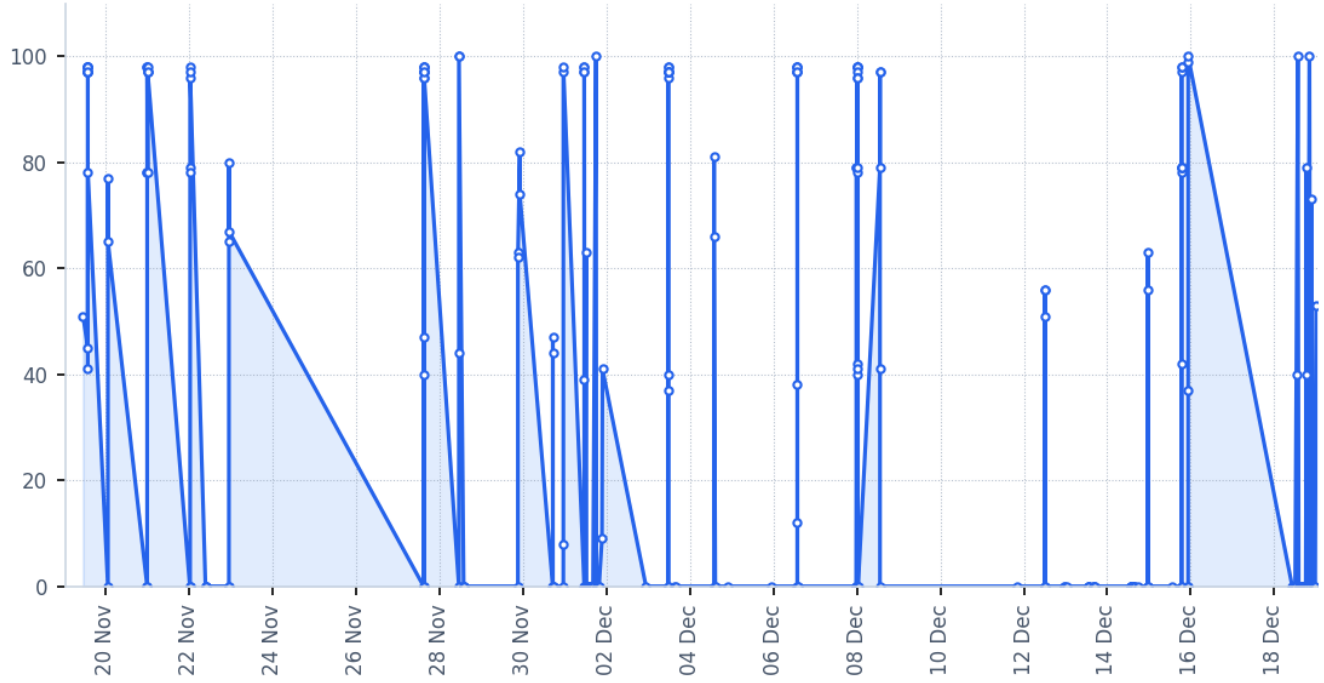
Repo ID	Name	Type
AppStream	Red Hat Enterprise Linux - AppStream	Local
BaseOS	Red Hat Enterprise Linux - BaseOS	Local

Analysis & Recommendations

- All systems healthy (<80%).

CPU Performance Analysis: redhatserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	53.0%	100.0%	38.4%	Transient Peak

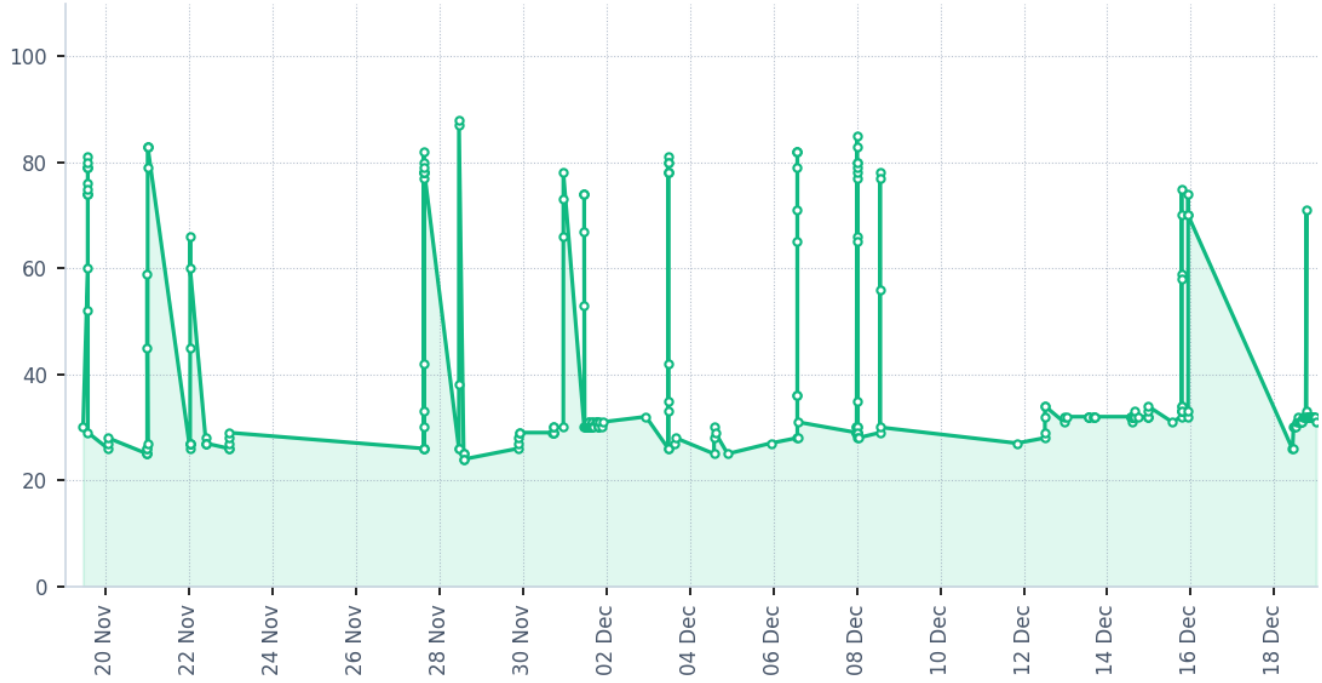
Analysis & Observations

Transient Spikes: While average load is normal (38.4%), critical peaks up to 100.0% were detected. Investigate specific scheduled tasks.

Data consistency check passed. No significant gaps detected in the monitoring period.

Memory Usage Analysis: redhatserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Metric	Current	Peak (30d)	Average	Assessment
Usage	31.0%	88.0%	42.1%	Normal
Swap	-	0.0%	-	Normal

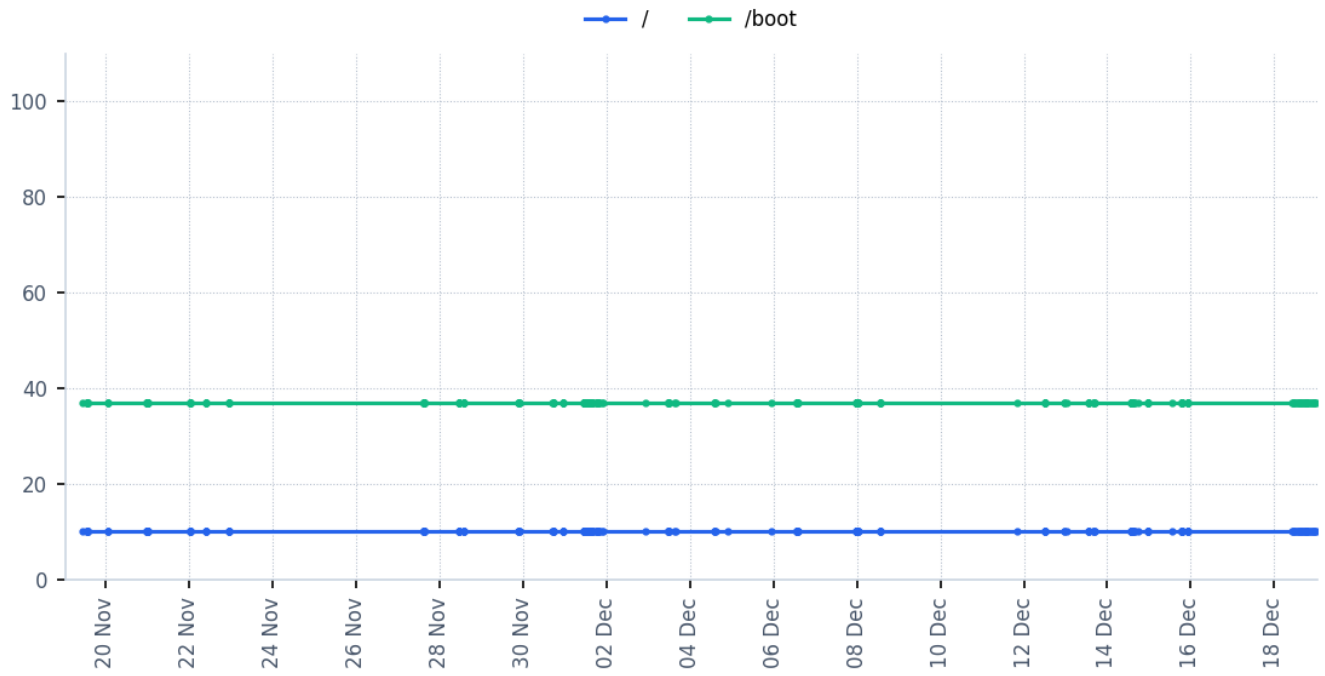
Analysis & Observations

Normal Operation: Resource consumption is within standard baselines (Avg: 42.1%).

Data consistency check passed. No significant gaps detected in the monitoring period.

Disk Usage Analysis (OS Partitions): redhatserver1.local

Scope: 19 Nov 2025 - 19 Dec 2025



Statistics Summary

Mount Point	Size	Current	Peak (30d)	Assessment
/	46G	10.0%	10.0%	Normal Operation
/boot	960M	37.0%	37.0%	Normal Operation

Analysis & Observations

Disk usage analysis is detailed in the table above based on OS partition monitoring.

APPENDIX: METHODOLOGY & CRITERIA

1. Health Score Interpretation

Score Range	Level	Meaning
90% - 100%	Excellent	All systems are healthy or have minimal load.
70% - 89%	Good / Fair	System operational but monitor warning signs.
50% - 69%	Warning	Multiple hosts are degraded or critical resources detected.
0% - 49%	Critical	Major outages or widespread failures. Immediate action required.

2. Status Definitions & Logic

Status	Weight	Condition / Logic
Up	10	Normal Operation. Online (< 6h). Resource Usage (CPU, MEM, DISK (/ root)) < 80%.
Warning	7	High Load. Online. Resource Usage (CPU, MEM, DISK (/ root)) 80% - 85%.
Degraded	3	Critical Load. Online. Resource Usage (CPU, MEM, DISK (/ root)) >= 86%.
Offline	0	Connection Lost. No data > 6 Hours (Last Seen).

3. Calculation Trace (Current Report)

$$\textbf{Health Score} = \frac{\text{SUM (Count x Weight)}}{\text{Total Hosts x 10}} \textbf{ x 100}$$

Current Data:	Total: 3 | Up: 2 | Warn: 0 | Deg: 1 | Off: 0
Step 1: Weighted Sum	(2x10) + (0x7) + (1x3) + (0x0) = **23**
Step 2: Max Score	3 Hosts x 10 Points = **30**
Step 3: Final Calculation	(23 / 30) * 100 = **76% (Good)**

4. Standards Alignment & Compliance Support

This document is generated to serve as verifiable operational evidence, designed to support the organization's compliance with international standards regarding IT service management and information security.

Metric Framework & Methodology:
The Health Score is calculated based on **Apdex (Application Performance Index)** principles for performance scoring, aligned with **NIST SP 800-137** guidelines for Information Security Continuous Monitoring (ISCM) and metric quantification.

ISO/IEC 20000-1:2018 (IT Service Management):
- **Clause 9.2 (Service Reporting):** Provides comprehensive reports on service performance and suitability.
- **Clause 8.7 (Service Availability):** Monitoring availability and recording outage events (Offline status).

ISO/IEC 27001:2022 (Information Security):
- **Control 5.37 (Documented Operating Procedures):** Serves as a record of routine preventive maintenance.
- **Control 8.6 (Capacity Management):** Monitoring resource utilization (CPU, Memory, Disk) to prevent system overload and ensure required system performance (formerly Control A.12.1.3 in the 2013 version).

5. DOCUMENT SIGN-OFF

Reference Service Report ID: [_____]

Verified By (Service Provider)	**Acknowledged By (Customer)**
_____ (Signature) **Name:** _____ **Position:** _____ **Company:** [PTS] Professional Team Services **Date:** _____ / _____ / _____	_____ (Signature) **Name:** _____ **Position:** _____ **Company:** Bank of Dream Public Company Limited **Date:** _____ / _____ / _____

Note: This report is a technical attachment to the maintenance service reference above.